

Melaku Alehegn

Data Engineer

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Addis Ababa, Ethiopia

SUMMARY

Data Engineer with a robust background in data management, ETL processes, and machine learning integration. Proficient in Python for data manipulation and modeling, SQL for database management, and leveraging ML frameworks. Experienced in optimizing data pipelines for efficiency and scalability, with a strong focus on data visualization and reporting tools. Skilled in designing and implementing scalable solutions for data storage and retrieval, complemented by a solid understanding of cloud platforms like AWS and Azure. Actively contributes to open-source projects, demonstrating a commitment to innovation and collaborative problem-solving.

EXPERIENCE

Data Engineer Intern

04/2024 - 07/2024 California, USA
10Academy

- Developed and optimized data pipelines for Retrieval-Augmented Generation (RAG), enhancing data retrieval and processing efficiency.
- Engineered and fine-tuned Large Language Models (LLMs) to improve model accuracy and performance in data-driven applications.
- Applied prompt engineering techniques to refine model outputs, ensuring high-quality data insights and decision-making.
- Collaborated effectively in cross-functional teams, leveraging strong communication and teamwork skills to deliver complex data engineering projects.

Opensource Python Developer Intern

12/2023 - 03/2024 Remote
OpenStack

- Enhanced the user interface of the OpenStack Manila project by implementing a streamlined workflow using Django views, improving the share network creation process. Achieved a 90% unit test coverage in the manila-ui project, ensuring robust code quality and reliability.
- Introduced Pre-commit and tox configurations in manila-ui, simplifying code formatting and standard adherence, facilitating smoother development workflows for team members.
- Collaborated closely with core reviewers in OpenStack to merge patches within release cycle timelines, ensuring timely integration of enhancements and fixes to maintain project momentum and quality.

Full stack Developer Intern

08/2023 - 12/2023 Addis Ababa, Ethiopia
MMCY TECH

- Developed several projects using Tailwind CSS and Vue.js, enhancing their design and functionality.
- Led the development of the job portal in the company's Applicant Tracking System (ATS), resulting in a 40% reduction in labor requirements, optimizing the recruitment process.
- Implemented strategic improvements to the ATS, particularly in the job portal, reducing time-to-hire by 50% and enhancing recruitment efficiency.
- Recognized with the Best Project Presenter Award for delivering compelling presentations, demonstrating outstanding communication skills and a captivating presentation style.

SKILLS

Technical Skills

Data Pipelines		ETL Processes	
Data Modeling		Data Warehousing	
SQL Databases(PostgreSQL, MySQL)			
NoSQL Databases (MongoDB)			
Data Cleansing and Transformation			
Cloud Platforms (AWS, Azure)			
Big Data Technologies (Kafka, Airflow)			
Python		CI/CD Pipelines	
React		Vuejs	
		Git & Github	

EDUCATION

ALX Software Engineering Program

ALX Africa
02/2022 - 02/2023 Nairobi Kenya
Python, SQL, C, DevOps

Cloud Developer Nanodegree

Udacity
05/2022 - 08/2022 Remote
Amazon elastic compute cloud, Cloud environment configuration, AWS IAM, AWS database services, Cloud security in AWS, AWS elastic beanstalk, Serverless development

Electrical and Computer Engineering

Addis Ababa Science & Technology University
09/2016 - 09/2021 Addis Ababa, Ethiopia
Database Design, Data Structures, Statistics and Probability, Object Oriented Programming (OOP)

PROJECTS

Prompt Generation System

A Prompt Generation system for generating prompts based on user descriptions, scenarios, and expected outputs, with algorithms and evaluation metrics ensuring alignment.

Data Warehouse For LLM Fine-tuning

Enhanced Amharic NLP by collecting, processing, and storing text and audio data from diverse online sources.